AI FOR STUDENTS

Artificial Intelligence

• Think • Create • Build with AI

Navneet Kumar

Welcome to Your AI Adventure! 🚀

Have you ever wondered how **ChatGPT** can answer your questions, how your **phone recognizes your face**, how **AI can create a picture** from a few words, or how a computer can learn from examples?

You are already living in a world powered by Artificial Intelligence.
But AI is not just about robots and computers. It is a powerful technology that can help us **learn, create, solve problems and explore new ideas**.

Throughout this book, you **won't just read about AI**.

You'll ask it questions, experiment with prompts, create images, make videos and music, build a game, investigate AI mistakes, and imagine the future.

A Note for Parents & Teachers

Artificial Intelligence is no longer a technology of the future. **It is already a part of our present,** and it is evolving every day.

As AI continues to grow and become a part of education, work, communication, and everyday life, it is important that **children understand this technology rather than remain disconnected from it.**

This book is designed to help bridge the **gap between the rapid growth of technology and the slower pace of formal education.**

Throughout this book, children will learn what Artificial Intelligence actually is, how it works, how it is changing the world, and how it may shape their future.

But there is an important goal behind this book:
We do not want children to become dependent on AI. We want them to learn how to think with it.

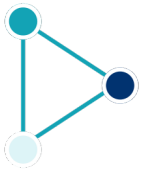
AI should be used as a tool for **learning, creativity, problem-solving, and exploration**—not as a replacement for curiosity, effort, or independent thinking.

We encourage parents and teachers to explore these activities with children, ask questions, discuss the answers AI provides, and remind them to check information, protect their privacy, and use technology responsibly.

We have made every effort to make this book clear and accurate, but there is always room for improvement. Your feedback can help us make it better.

For feedback, suggestions, or concerns:
Email: **Navneetkyadav40@gmail.com**

Table Of Content



Chapters Page

1.What is AI..... 07-16

2.AI All Around Us..... 17-27

3.How Does AI learns..... 28-39

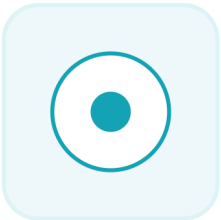
4.Talking To AI: Meet LLMs..... 40-51

5.Using AI For Learning & Home Work..... 52-62

6.Using AI safely & Responsibly..... 63-74

7.Fun With AI + Mini Projects..... 75-87

8.AI & The Future..... 88-96

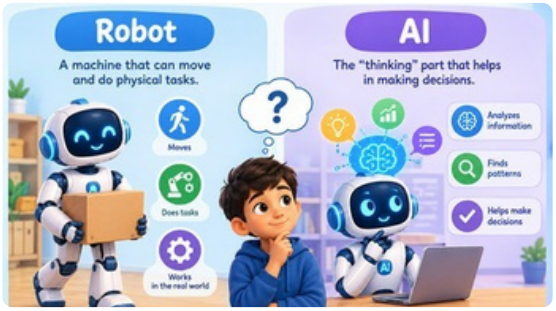


Is AI the Same as a Robot ?

Many children think AI means robots.

This is only partly true.

- A **robot is a machine** that can move and do physical tasks.
- **AI is the "thinking"** part that helps in making decisions.



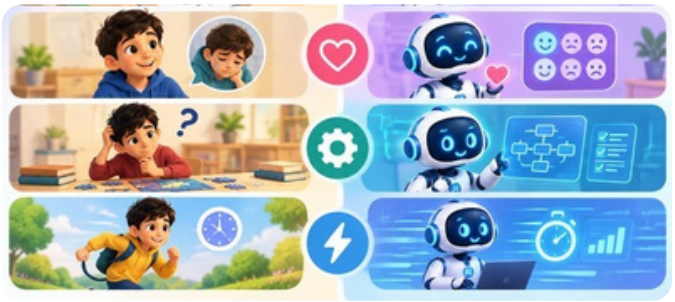
Most AI (like ChatGPT, Gemini, YouTube recommendations) does **not live inside a robot**. It lives quietly inside apps and websites.

THE SIMPLE IDEA

AI is like the brain.
The phone, app, or robot is like the body.
You can have a smart brain without a moving body.

Let's Compare Humans and AI

Humans and AI can **both learn, solve problems**, and make decisions, but they do these things in very different ways.



Human intelligence comes from real-life experiences and feelings, while Artificial Intelligence (AI) comes from computer code and math patterns

- **Humans**: Learn by living, making mistakes, and trying again (like falling off a bicycle and learning to balance).
- **AI**: Learns by looking at millions of examples and data fed by humans (like studying thousands of pictures of cats)



Think About It

Imagine you and an AI are both asked:
"**Design a dream playground for your school.**"

You might think about what your friends **enjoy, memories from playing, safety**, and what would make everyone happy.
AI can suggest many playground ideas by **finding patterns from the information it has learned.**
Both can come up with ideas—but they do it in different ways.

Humans vs AI

Human	AI
Learns from experiences	Learns patterns from data
Can use common sense	Can struggle with common sense
Has emotions	Doesn't have human emotions
Can understand context deeply	Can misunderstand context
Can be creative	Can generate creative-looking content
Can make judgments	Can make predictions/recommendations



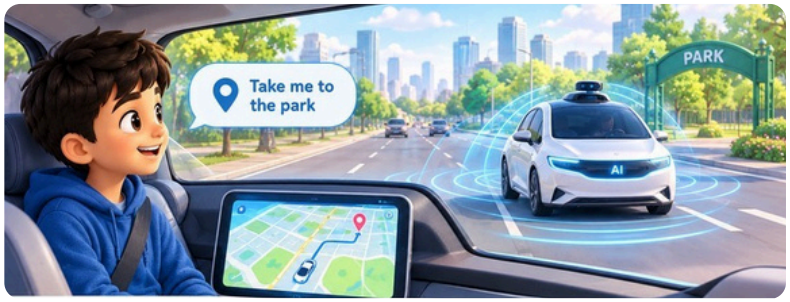
Key Idea

AI can be powerful, but it is not a human brain. Humans bring experiences, **emotions, values**, and understanding of the real world that AI does not experience in the same way.

AI is not simply a smarter version of a human. It is a different kind of technology with different strengths and limitations.

Driverless cars (AI in cars)

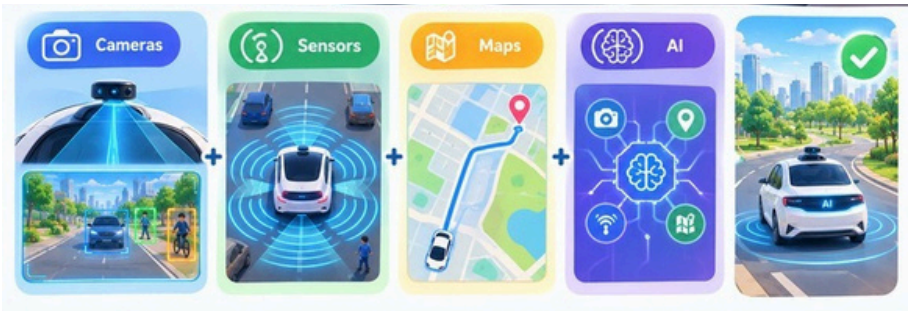
Imaginegettinginto acar, tellingitwhere you want to go, and the car drives there without a person controlling the steering wheel or pedals!



These cars use **AI, cameras, radar, lidar, sensors, maps**, and powerful computers to understand the road and decide how to drive.

A driverless car can be designed to:

- 🚗 Detect other vehicles
- 👤 Detect pedestrians
- 🚦 Recognize traffic lights and road signs
- 🛣️ Identify road lanes
- 🧭 Follow a route to a destination
- 🛑 Slow down or stop when something is in its path



Simple Example

Imagine a pedestrian suddenly crosses the road.

Camera & sensors → AI detects the person → system predicts what is happening → car slows down or stops

AI helps the car understand what is happening around it and choose an appropriate action.

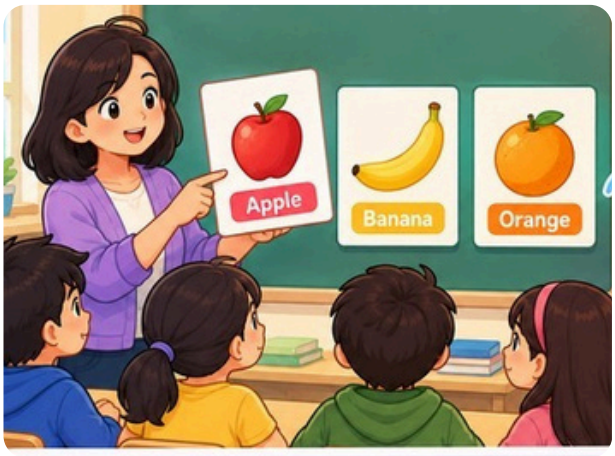
Three Ways AI Can Learn

Just like humans can learn in different ways, AI systems can also learn in different ways. Three important approaches are **Supervised Learning, Unsupervised Learning, and Reinforcement Learning**.

1. **Supervised Learning** — Learning With Answers

Imagine your teacher shows you pictures and tells you the answers:

🍎 → Apple
🍌 → Banana
🍊 → Orange



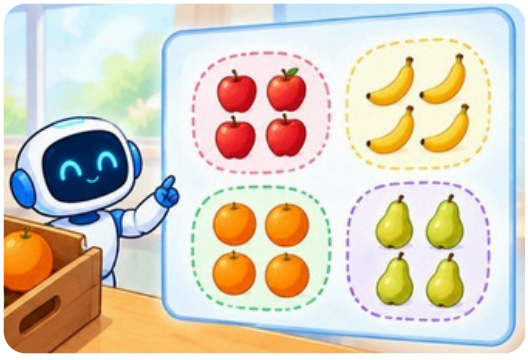
You learn by looking at examples where the correct answer is already provided. AI can learn in a similar way. It is given labeled training data and learns patterns that help it **predict the correct answer** for new data.

Examples + Correct Answers → Learn → Predict

2. **Unsupervised Learning** — Finding Patterns on Its Own

Now imagine your teacher gives you a box full of fruits but doesn't tell you their names.

You might notice:
🍎 🍎 🍎 → These look similar
🍌 🍌 🍌 → These look similar
🍊 🍊 🍊 → These look similar



You have found groups based on patterns **without being given the answers.**

AI can do something similar. It can examine data and find patterns, groups, or similarities **without labeled answers.**

Data → Find Patterns → Groups or Discoveries

What is a Prompt?

A prompt is the **instruction or question** you give to an AI tool (like ChatGPT, Gemini, or any other AI).



It could be:

A Question :	**Why is the sky blue?**
An Instruction :	**Write a short story about a robot.**
A Task :	**Give me five quiz questions about the solar system.**
A Combination :	**Explain photosynthesis to a Class 6 student using a simple real-life example and then give me five quiz questions.**

Prompt Playground

Now it's time to become a **Prompt Explorer!**
Try these experiments and compare the answers.

You don't need special coding skills to write a good prompt. You just need to learn how to give AI clear instructions.

Experiment 1: Be More Specific

Start with a simple prompt:
Tell me about dogs.
Now try:
Tell me 5 interesting facts about dogs for a 10-year-old. Use simple language.

Experiment 2: Change Who AI Is Talking To

Use the same topic but change the audience.

Prompt A :***Explain gravity.***
Prompt B : ***Explain gravity to a 13 years old.***
Prompt C : ***Explain gravity to a 5 year old.***

The topic is the same, but the answers may be very different.
Lesson: Tell AI who **the answer is for.**

THE AI TRAFFIC LIGHT

🟢 GREEN — USE AI

Examples:

- brainstorming
- explanations
- practice questions
- creative ideas
- organizing information

🟡 YELLOW — PAUSE & CHECK

Examples:

- school research
- factual claims
- important information
- AI-generated images
- unfamiliar sources

🔴 RED — STOP

Examples:

- passwords
- OTPs
- private information
- impersonating someone
- cheating
- harmful content
- sharing fake information

ACTIVITY 4 — AI TRAFFIC LIGHT

classify each situation:

🟢 **GREEN — Generally okay**
🟡 **YELLOW — Stop and think**
🔴 **RED — Don't do it**

1. “Explain photosynthesis to me.” 🟢 / 🟡 / 🔴

2. “Here is my password. Help merememberit.” 🟢 / 🟡 / 🔴

3. “Give me five ideas for a scienceproject.” 🟢 / 🟡 / 🔴

4. “Here is my friend's private photograph.Makeitfunny.” 🟢 / 🟡 / 🔴

5. “Give me practice questions for my mathsexam.” 🟢 / 🟡 / 🔴

6. “Here is my OTP. Tell me what it means.” 🟢 / 🟡 / 🔴

7. “Create an imaginary picture of a robot visitingMars.” 🟢 / 🟡 / 🔴

Chapter

8

AI & The Future



One day, Aarav looked up at the sky and wondered,
"What will the world be like when I grow up?"

He pictured **smart cities** that never sleep, **friendly robots** that help people, **driverless cars** all around wild new inventions, and jobs that don't even exist yet.

He thought "This means we dont have to work anymore everything will be done by AI"

Aarav smiled.
"The future is going to be interesting..."



What You Will Learn Today

- Explore different types of creative AI tools (beyond just chat)
- Create something original using AI as a helper
- Try simple hands-on AI projects
- Reflect on how AI can support creativity

AI And Environment

AI feels digital.

But AI systems run on physical computers.

Those computers need:

- ⚡ Electricity
- 💧 Cooling
- 🏢 Data centres
- 🖥️ Hardware

Large data centres can consume **substantial amounts of electricity** and, depending on **cooling systems and location, water.**

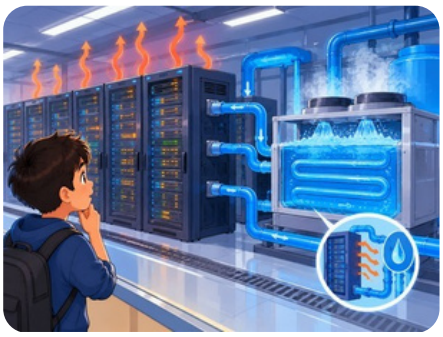
The International Energy Agency projects that electricity consumption from data centres will more than double by 2030, with AI as a major driver of the increase.

Why does AI need water

Thismay**surprise students.**

Computers produce heat.

Data centres need **cooling systems** to keep equipment from overheating.
Some cooling systems use water.



AI And Your City

When you go to school, look around you.



Do you sometimes see garbage lying on the roadside?



Have you noticed a pothole that has been there for weeks?



Maybe you have been stuck in traffic even when the road doesn't seem very busy.



During **heavy rain**, have you seen water collecting on roads because **drains are blocked?**

These are not science-fiction problems. They are problems that many cities already face.

Now imagine using *AI to help solve them.*

1. Managing Waste



Normally, someone needs to **notice the problem** and **report** it.
In the future, a **smart camera or sensor could monitor** waste bins.
When a bin becomes full, AI could detect it and automatically alert the waste collection team.

2. Finding Potholes



AI could analyse the images and **detect possible pot holes**.
It could then record:

- 📍 Where the pothole is
- 📏 How large it appears
- ⚠️ How serious the damage may be

The information could be **sent to the road maintenance team.**

3. Reducing Traffic



One road may be crowded while another nearby road is almost empty.
AI can analyse traffic information from **cameras, sensors, maps, and other systems** to understand traffic patterns.
It could help cities:

- Detect traffic jams
- Predict busy periods
- Improve traffic signal timing
- Suggest alternative routes
- Identify accident-related delays



4. Preventing Waterlogging

AI could combine information from:

- 🌧️ Weather forecasts
- 💧 Water-level sensors
- 🕳️ Drainage systems
- 📍 Maps of low-lying areas



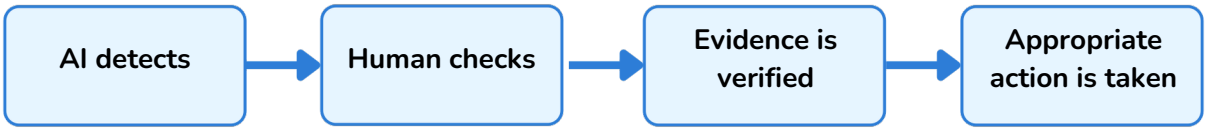
A city could then prepare before heavy rainfall arrives.

5. Can AI Detect People Who Pollute?

Now imagine something even more advanced.

Suppose a **city has cameras in areas** where people frequently throw garbage on the road.
AI could potentially **detect someone throwing garbage in a restricted area** using cameras. A possible system could be:





Human + AI

AI is becoming more powerful, but **the future is not just about AI.** It is about how **humans and AI** work together.

AI can:

- Find patterns in huge amounts of data
- Generate ideas and content
- Automate repetitive tasks
- Help us solve problems
- Make predictions

But humans can:

- Think critically
- Make important decisions
- Understand emotions
- Show empathy
- Decide what is right and fair
- Take responsibility

Think of AI as a powerful tool not the replacement for human intelligence.